Advanced Combustion via Microgravity Experiments (ACME)

ACME is focused on advanced combustion technology via fundamental microgravity research. The primary goal is to improve efficiency and reduce pollutant emission in practical terrestrial combustion. A secondary objective is fire prevention, especially for spacecraft.

ACME included five independent experiments (see ACME Experiments below) investigating laminar, gaseous, non-premixed flames.

The ACME experiments will be conducted with a single modular set of hardware (see ACME Implementation) in the Combustion Integrated Rack (CIR) on the International Space Station (ISS).

An ACME precursor, Structure & Liftoff In Combustion Experiment (SLICE), was conducted in the ISS' Microgravity Science Glovebox (MSG) in early 2012.

The ACME design is complete and the engineering hardware is being assembled for integrated testing. On-orbit testing is expected to begin in 2017 and continue for a few years.

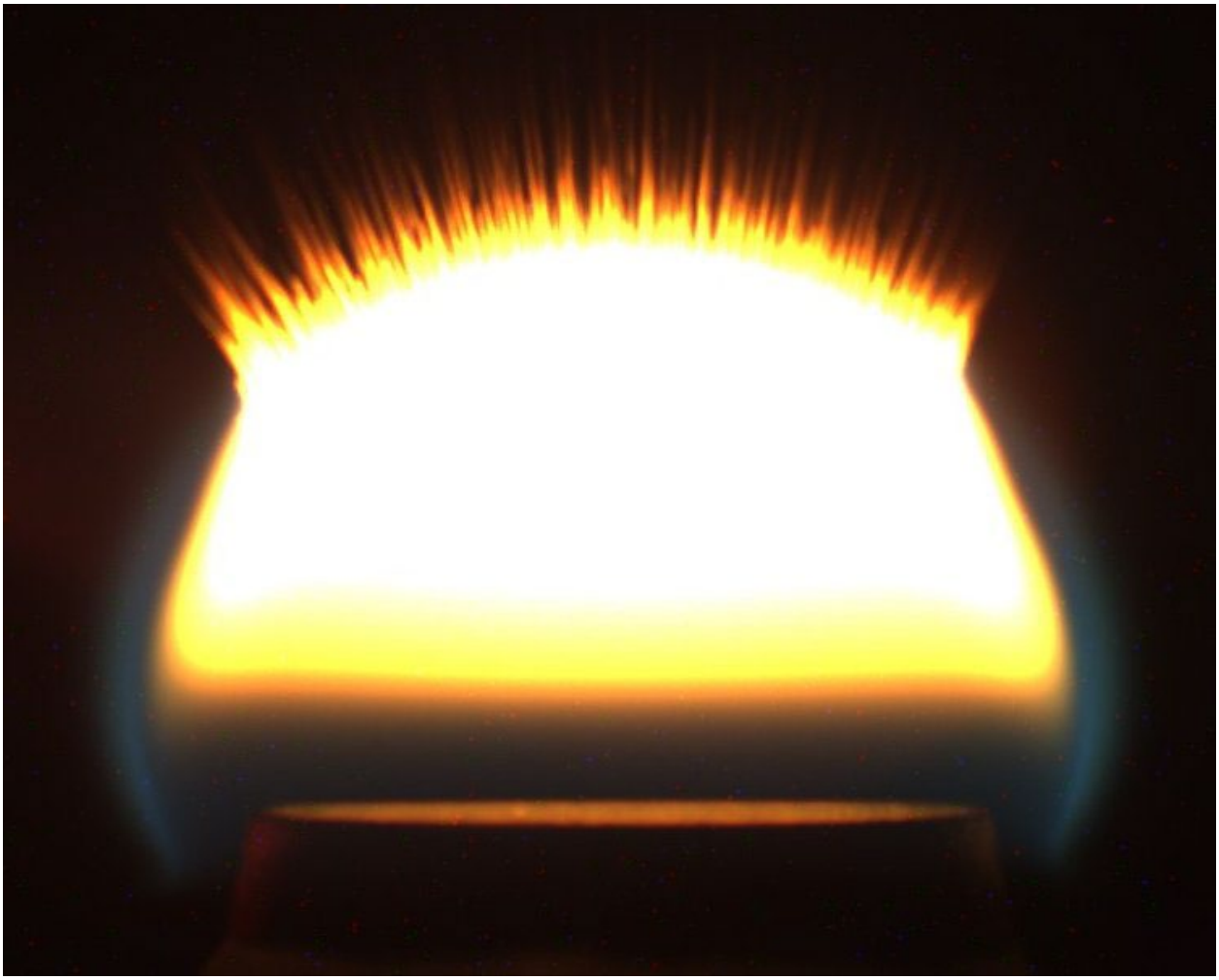
ACME Status

For the latest updates and photos,
see www.facebook.com/space.flames/ and www.flickr.com/photos/space-flames, respectively.

ACME Experiments

- The Advanced Combustion via Microgravity Experiments (ACME) project includes six independent experiments investigating laminar gaseous non-premixed flames. In other words, the flow is smooth and without vortices, the fuel is a gas (and not a liquid or solid), and the fuel and oxygen are not mixed in the burner (but are instead on opposite sides of the flame sheet).
- ACME is focused on advanced combustion technology via fundamental microgravity research. The primary goal is to improve efficiency and reduce pollutant emission in practical terrestrial combustion, for example through the development and verification of improved computational models. A secondary objective is fire prevention, especially for spacecraft.
- In addition to enhanced performance, the improved modeling capability resulting from ACME could lead to reductions in the time and cost for combustor design.
- Some specific ACME goals are to improve our understanding of combustion at fuel lean conditions where both optimum performance and low emissions can be achieved, soot control and reduction, oxygen-enriched combustion which could enable practical carbon sequestration, flame stability and extinction limits, and the use of electric fields for combustion control.
- The goal of ACME's spacecraft fire prevention research is to improve our fundamental understanding of materials flammability, such as extinction behavior and the conditions needed for sustained combustion, and to assess the relevance of existing flammability test methods for the screening and selection of materials for spacecraft.

Burning Rate Emulator (BRE)



Unlike the other current ACME experiments, the Burning Rate Emulator (BRE) experiment is focused on fire prevention, especially in spacecraft. Specifically, BRE's objective is to improve our fundamental understanding of materials flammability, such as ignition and extinction behavior, and assess the relevance of existing flammability test methods for low and partial-gravity environments. The burning of solid and liquid fuels will be simulated by using a flat porous burner fed with gaseous fuel. The fuel flow rate will be controlled based on the measured heat flux (at the burner) and surface temperature, mimicking the dependence of condensed-phase fuel vaporization on thermal feedback. A small number of gaseous fuels will be used to simulate the burning of fuels such as paper, plastic, and alcohol by matching properties such as the surface temperature and smoke production.

Coflow Laminar Diffusion Flame (CLD Flame)

Research, especially including that already conducted in microgravity, has revealed that our current predictive ability is significantly lacking for flames at the extremes of fuel dilution, namely for sooty pure-fuel flames and dilute flames that are near extinction. The general goal of the Coflow Laminar Diffusion Flame (CLD Flame) experiment is to extend the range of flame conditions that can be accurately predicted by developing and experimentally verifying chemical kinetic and soot formation submodels. The dependence of normal coflow flames on injection velocity and fuel dilution will be carefully

examined for flames at both very dilute and highly sooting conditions. Measurements will be made of the structure of diluted methane and ethylene flames in an air coflow. Lifted flames will be used as the basis for the research to avoid flame dependence on heat loss to the burner. The results of this experiment will be directly applicable to practical combustion issues such as turbulent combustion, ignition, flame stability, and more.



GENERAL OVERVIEW

- **EXPERIMENT:** The Coflow Laminar Diffusion Flame (CLD Flame) experiment investigates combustion at the extremes of the fuel dilution spectrum – namely pure-fuel flames that produce soot, and highly diluted flames that are close to extinction. At first glance, the pure-fuel flame shapes resemble familiar candle flames. However, comparison of the flames under normal gravity and microgravity conditions shows a striking difference (in microgravity, the absence of buoyancy means hot gases no longer rise).
- **SOOT:** The yellow color of these flames comes from the incandescence of hot soot, which is made up of tiny carbon particles. These soot particles have been shown to have adverse health effects as well as being a significant contributor to climate change. Sooty microgravity flames are wider and taller than their normal gravity counterparts, and the concentration of soot can be larger by nearly a factor of ten. By careful design of the burner and close monitoring of the flow conditions, the images from the flames produced can be analyzed to give quantitative information on temperature and soot concentration. This data can then be used to compare and improve computer models of soot formation and destruction, which will have benefits on earth as well as in space.
- **FUEL DILUTION:** When the fuel is diluted with an inert gas (nitrogen in our case), the soot disappears and the flame can lift off the burner surface and appear as a faint blue cone. With increased dilution, the flame becomes a flat disk, and finally extinguishes. Once again, the microgravity flames are seen to behave quite differently, with highly diluted fuels producing stable flames that would not be possible under normal gravity conditions. In addition to implications for fire safety in space, results from these weak flames are important for understanding their stabilization mechanism and dilution limits.

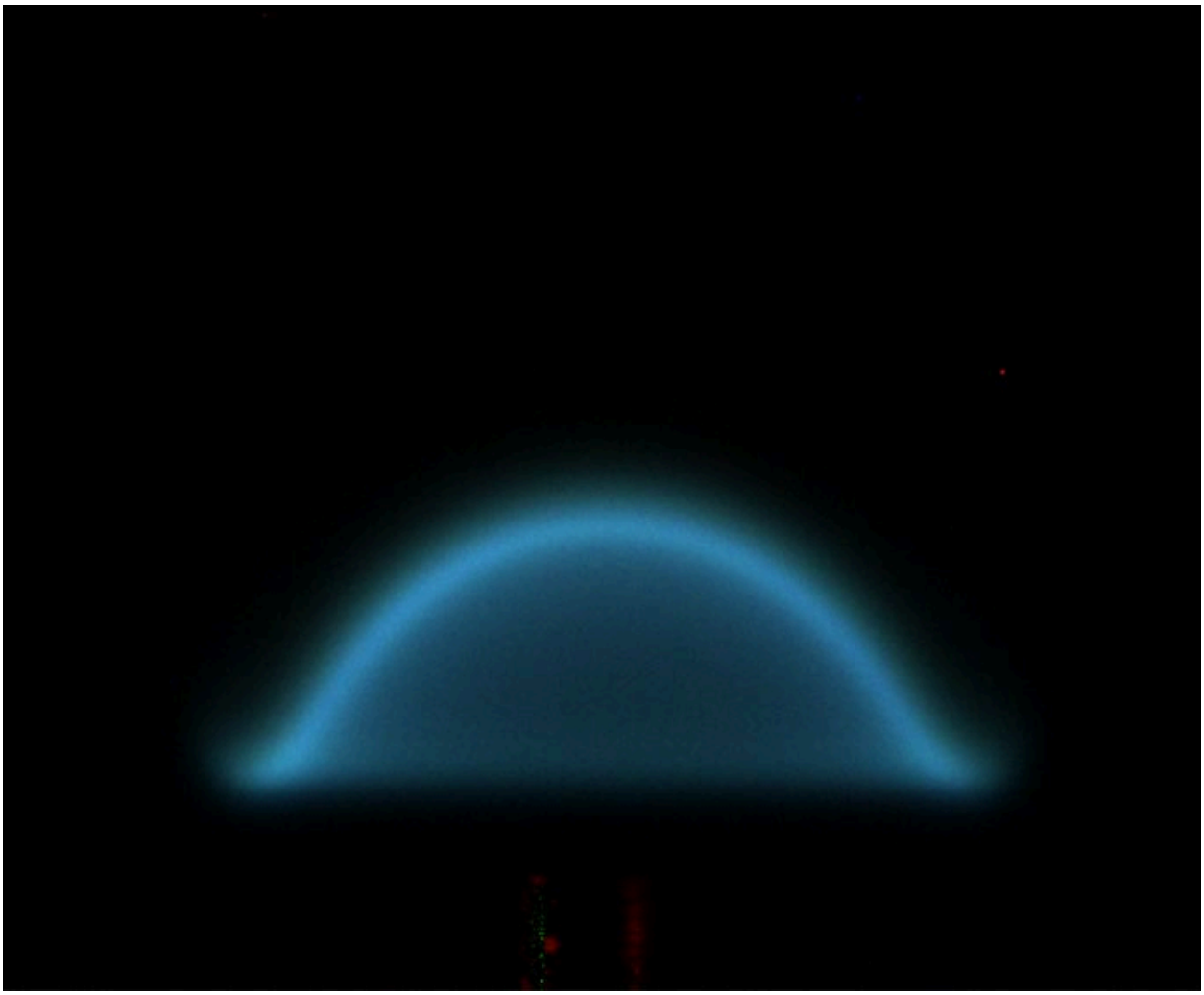
SELECTED FINDINGS

- For sooty flames, results show that the absence of gravity can have a major effect. Flames are taller, wider, cooler, and far sootier in microgravity as compared to normal gravity.

- Soot volume fraction measurements indicate that microgravity flames contain five to eight times more soot than the equivalent normal gravity flames.
- The data has shown that for highly diluted weak flames, microgravity flames are more stable than their normal gravity counterparts. The dilution limits are extended allowing for lifted flames at lower fuel concentrations than observed in normal gravity.

The experiment has provided high-fidelity quantitative measurements of axially symmetric non-premixed flames over a wide range of flow conditions, without the complication of buoyancy forces that are experienced by flames on Earth. These results, coupled with carefully measured flow boundary conditions, will provide a rigorous testbed for verifying computational models.

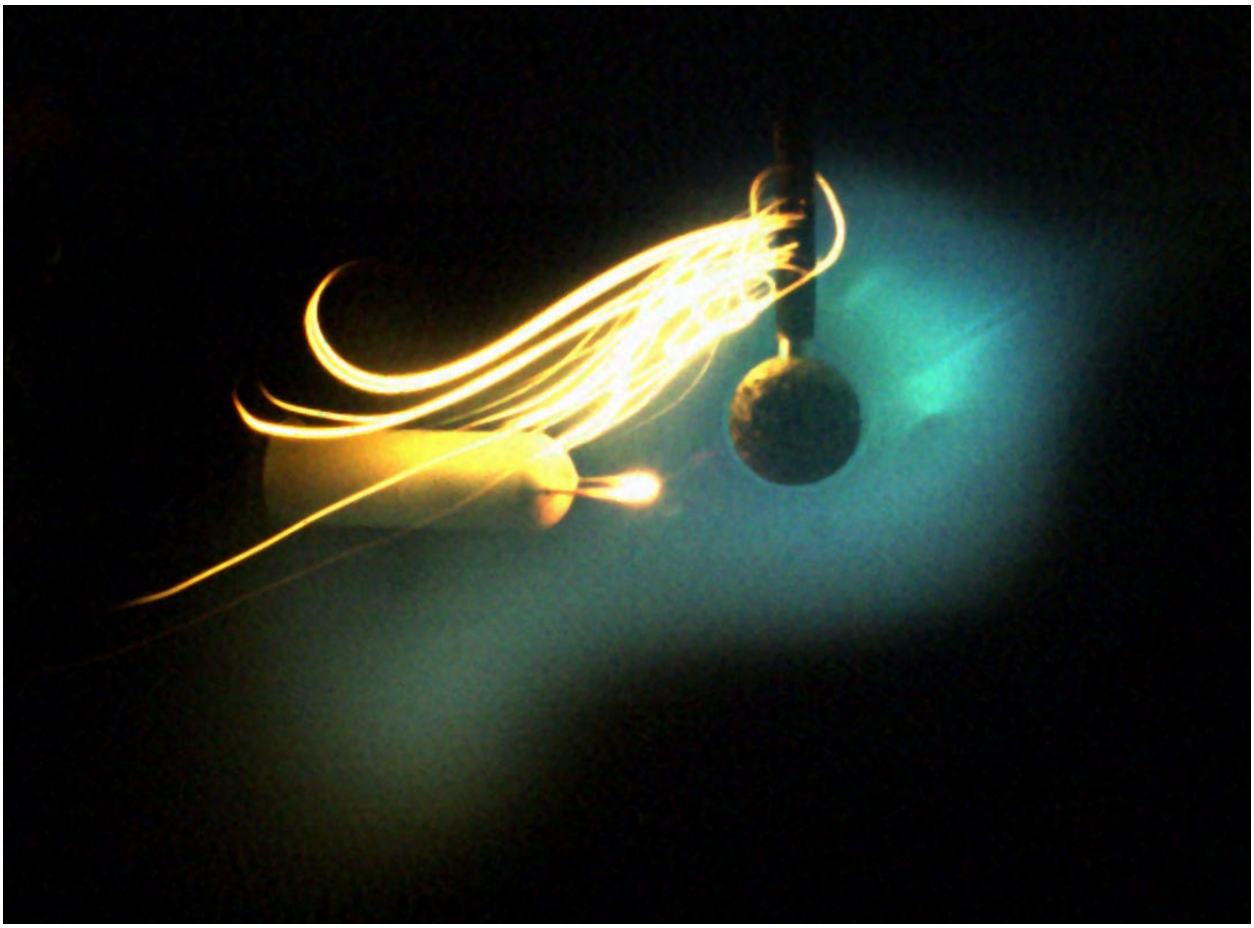
Electric-Field Effects on Laminar Diffusion Flames (E-FIELD Flames)



Electric fields can strongly influence flames because of its effect on the ions present as a result of the combustion reactions. The direct ion transport and the induced ion wind can modify the flame shape, alter the soot or flammability limits, direct heat transfer, and reduce pollutant emission. The purpose of the Electric-Field Effects on Laminar Diffusion Flames (E-FIELD Flames) experiment is to gain an improved understanding of flame ion production and investigate how the ions can be used to control non-

premixed flames. Outside reviewers recently concluded that the experiment "… will contribute to our critical understanding to our knowledge of combustion processes in the presence of electric fields." The experiment will be conducted with a normal coflow flame (as in the CLD Flame experiment) or perhaps with a simple gas-jet flame, where there is no surrounding coflow. An electric field will be generated by creating a high voltage (up to 10 kV) differential between the burner and a flat circular mesh suspended above (i.e., downstream of) the burner. Measurements, as a function of field strength and fuel dilution, will be made of the ion current through the flame and the flame's response time to electric forcing.

Flame Design



The primary goal of the Flame Design experiment is to improve our understanding of soot inception and control in order to enable the optimization of oxygen enriched combustion and the "design" of non-premixed flames that are both robust and soot free. An outside review panel recently declared that Flame Design "… could lead to greatly improved burner designs that are efficient and less polluting than current designs." Flame Design will investigate the soot inception and extinction limits of spherical microgravity flames, created in the same manner as for the s-Flame experiment. Tests will be conducted with various concentrations of both the injected fuel (i.e., ethylene or methane) and the oxygen enriched atmosphere in order to determine the role of the flame structure on soot inception. The effect of the flow direction on soot formation will be assessed with an inverse spherical flame unless such testing is not approved by the Payload Safety Review Panel (PSRP). If inverse spherical flame testing is not allowed, the plan is to use a coflow burner, conducting both normal and inverse flame tests. In the

case of the inverse flames, the oxygen/inert mixture is injected from a central tube, while the fuel is ejected from a surrounding annulus. The Flame Design experiment will explore whether the stoichiometric mixture fraction can characterize soot and flammability limits for non-premixed flames like the equivalence ratio serves as an indicator of those limits for premixed flames.

Structure and Response of Spherical Diffusion Flames (s-Flame)

The purpose of the Spherical Flame (s-Flame) experiment is to advance our ability to predict the structure and dynamics, including extinction, of both soot-free and sooty flames. The spherical flame, which is only possible in microgravity, will be created through use of a porous spherical burner from which a fuel/inert gas mixture will issue into the CIR chamber. Flames will be ignited at non-steady conditions and allowed to transition naturally toward extinction. Tests will be conducted with various inert diluents, in both the fuel and chamber atmosphere. The fuel gases include hydrogen and methane for soot-free flames, and ethylene for sooty flames. One experiment objective is to identify the extinction limits for both radiative and convective extinction (i.e., at high and low system Damkohler numbers, respectively). Another objective is to determine the existence, onset, and nature of pulsating instabilities that have been theoretically predicted to occur in such flames with fuel/diluent mixtures that are above a critical Lewis number.

https://youtu.be/f9AQ-OebO_U

Cool Flames Investigation with Gases (CFI-G)

Cool diffusion flames were discovered during droplet combustion experiments aboard the International Space Station (ISS) in 2012, and this initiated a rapidly growing field of combustion research. Droplet experiments necessarily involve unsteady burning rates, so cool diffusion flames have never been observed as steady spherical flames. A natural progression toward understanding cool diffusion flames is to observe quasi-steady spherical flames on porous burners, and this is the topic of this research. The flames to be studied are rich in the physics and chemistry of combustion and thermal transport processes.

This experiment was selected for spaceflight by the Center for the Advancement of Science in Space (CASIS) and the National Science Foundation (NSF).

GENERAL OVERVIEW

- **RELEVANCE:** Most internal combustion engines are designed using computer models that neglect cool flame chemistry, but ignition and flame propagation in engines depend on cool flame chemistry. Cool flame chemistry also has a significant impact on fuel octane and cetane numbers, whose understanding has large economic consequences.
- **MICROGRAVITY:** The flame system to be studied will allow steady spherical cool diffusion flames to be observed for the first time. The tests, performed with gaseous fuel and a porous spherical burner, cannot be performed in normal gravity or in ground-based microgravity facilities.
- **MODELING:** Observing cool flames burning as steady spherical diffusion flames will yield valuable insight into cool flame chemical kinetics. An advanced Computational Fluid Dynamics (CFD) model will simulate the flames and will aid the testing and development of cool flame kinetics models.

- **SHARING OF RESULTS:** The research will be disseminated widely to the combustion research community and to industry. It should advance the state-of-the-art in cool flame kinetics mechanisms, which in turn could lead to cleaner, more efficient internal combustion engines.

Benefits & Relevance

Why study combustion?

- In the United States, nearly 70% of our electrical energy is generated through the combustion of fossil fuels. For example, in 2012, electricity was generated in the U.S. by burning the following fuels, where the percentages indicate the fraction of the total U.S. electrical generation: coal (37%), natural gas (30%), biomass (1.4%), and petroleum (1%).
- Our transportation is heavily reliant on combustion, even for electric vehicles because most of our electricity is generated by combustion.
- With combustion, we heat our homes, water, food, etc. and also generate heat for industrial processes.
- Our reliance on imported fuel contributes to our national trade deficit and affects our national security.
- Combustion is a leading man-made source of greenhouse gases, where carbon dioxide is the most important example.
- Combustion is the primary man-made contributor to acid rain.
- Soot contributes to global warming and is a health problem.
- Fire safety!
- Given our pervasive use of combustion as an energy source, the U.S. consumes fossil fuels which cost on the order of a trillion dollars annually. Therefore, even small improvements in combustion efficiency would significantly reduce fuel needs and pollutant production.

Why study combustion in microgravity?

- Flames are strongly affected by gravity because the high-temperature combustion gases are much less dense than the cooler atmosphere which surrounds the flame. Gravity pulls more forcefully on the denser atmosphere and the hot gases are pushed upward as a result. This gravity-driven upward motion of material that is less dense that the surrounding fluid is referred to as buoyancy. This gravity-driven motion, referred to as buoyant convection, feeds the flame with fresh reactant – normally oxygen (in the air) – and removes the combustion products (e.g., carbon dioxide and water vapor) from the flame vicinity.
- Low-momentum flames are dramatically influenced by the effective elimination of buoyant convection, where the resulting effects are often advantageous for analysis.
- Spherically symmetric flames can be created enabling one-dimensional analysis. Two of the current ACME experiments take advantage of this feature.

- Flicker, which is a buoyancy-driven (i.e., gravity-driven) instability, is eliminated yielding quasi-steady flames.
- Length scales are increased in microgravity flames facilitating analysis of the flame structure.
- Momentum-dominated flames, which are important for most practical combustion, can be studied at low velocities to simplify analysis.
- Microgravity flames tend to have a much stronger sensitivity to their atmosphere and exhibit a much broader range of characteristics than normal-gravity flames because of the near absence of buoyant entrainment.
- The long residence times in microgravity flames can lead to strong soot production, but many microgravity flames are soot free.
- Microgravity flames are great for studies of limit and stability behavior where chemical kinetics are important. Soot, extinction, and stability limits are being studied in the ACME experiments.
- Microgravity is of course the appropriate environment for studies related to spacecraft fire safety.

Why study combustion on the ISS?

- Microgravity durations in drop facilities, such as the 2.2 Second Drop Tower and 5.18-second Zero Gravity Research Facility, are (1) too short for soot to achieve quasi-steady conditions, and (2) too short to establish a flame and then vary its flow rate, for example, to investigate stability or extinction limits.
- While research aircraft flying in parabolic maneuvers can provide reduced gravity durations of ~20 seconds, low-momentum flames are often dramatically disturbed by the aircraft vibrations. Although the jitter can be avoided if the experiment is floated within the aircraft, that reduces the low-gravity duration to mere seconds.

Publications

Link to PDF with all publications listed.

Downloads

A collection of downloads/documents related the ACME.

Need to upload all documents from old site.

ACME related Links

- ACME Image Processing Software (OMA2 and QtOma)
- Combustion Integrated Rack (CIR, facility in which ACME will be conducted)
- ISS Researchers Guide to Combustion Science
- International Space Station (ISS)

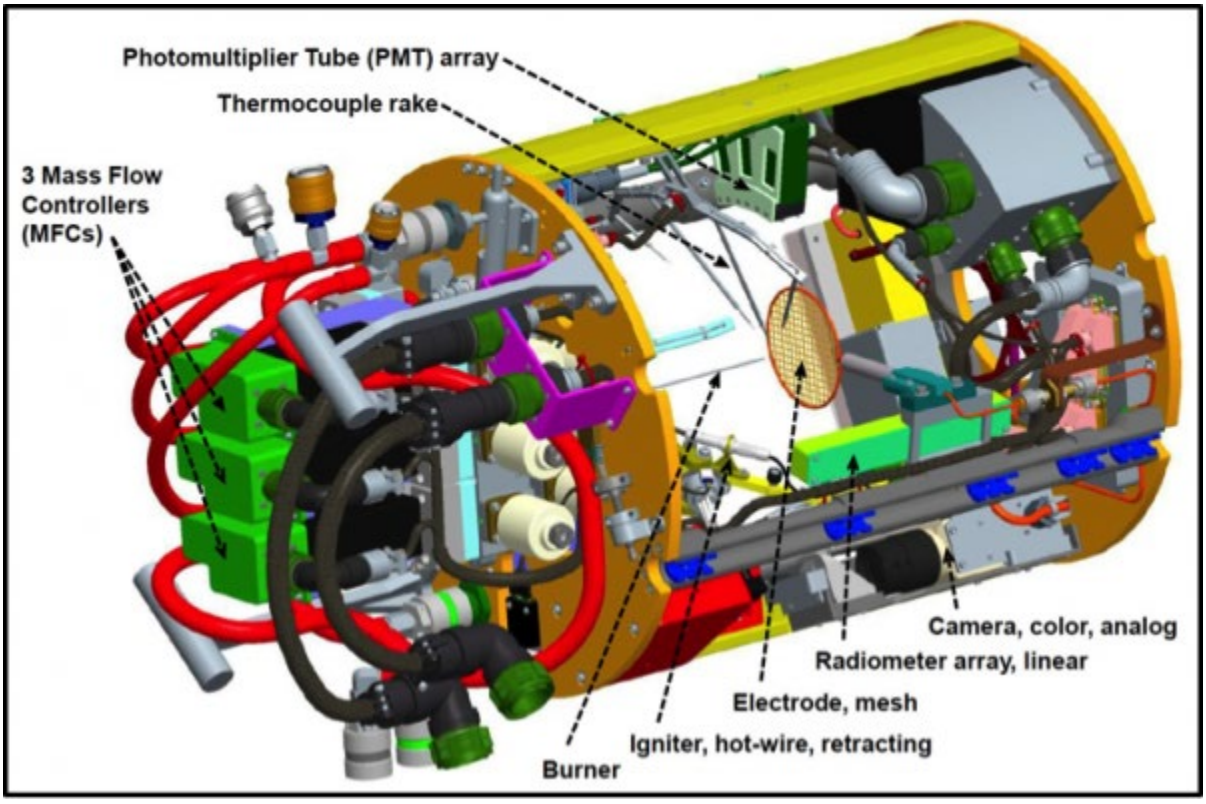
- NASA Glenn Research Center (lead center for ISS combustion research)
- Profs to a Flame (cover story, pp. 24-29)
- Structure & Liftoff In Combustion Experiment (SLICE, an ACME precursor)
- To Understand Combustion, Yale Engineers Look to the Skies
- ZIN Technologies, Inc. (ACME payload developer)

Implementation

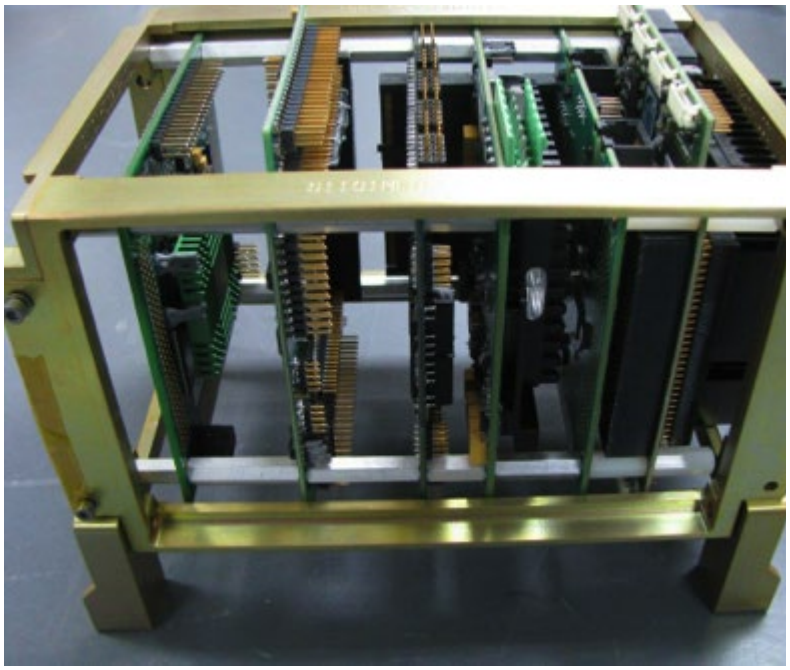
The ACME experiments will be conducted with a single modular set of hardware in the Combustion Integrated Rack (CIR) on the International Space Station (ISS), where the CIR is depicted below.



The three main elements of the ACME-unique hardware are the chamber insert (shown below), a high-definition color camera, and an avionics package (for data acquisition and control and also depicted below).







Engineering model of the ACME avionics package, with the internal card stack shown at the right (where the two photographs are at different scales).

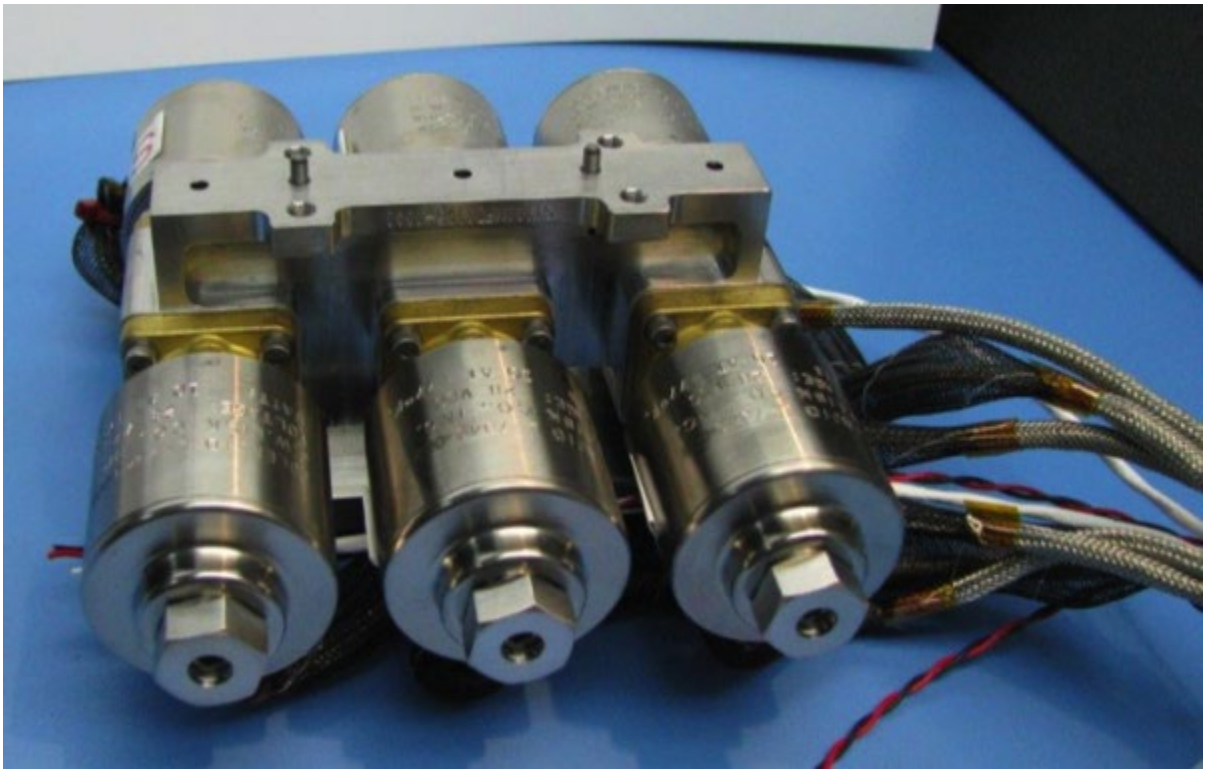
Burners and Igniter

- ACME provides common mounting (on the chamber axis) of a variety of burners, which are depicted and described below.
- Most burners will be equipped with a surface thermocouple, and each burner for the BRE experiment will have two heat flux transducers.
- The internal pressure of the porous burners will be measured to allow monitoring of the pressure differential across the outlet surface.
- The retracting and resistively heated igniter will have a crew-exchangeable tip and arm (to accommodate the various burners) and its stepper-motor drive will allow for precise positioning.



Gas Delivery

- ACME's chamber insert includes three mass flow controllers for the delivery of fuel, oxidizer or inert (e.g., helium or carbon dioxide), and nitrogen to the burner or for chamber filling.
- The maximum fuel flow is 2 slpm (on a nitrogen basis), whereas much higher flows, e.g., 10 slpm (on a nitrogen basis), are possible for the oxidizer and nitrogen lines. The crew can exchange the mass flow controllers to provide appropriate ranges that maximize the flow accuracy.
- The gas delivery system enables on-orbit dilution of the fuel or oxidizer (but not both simultaneously), coflow flames, inverse flames (e.g., oxidizer delivery to the center of a coflow burner), and premixed flames, although the latter are not planned for any of the current ACME experiments.
- Gases can also be delivered premixed to the space station where that is necessary in some circumstances, such as the blending of different fuel gases.
- CIR's standard bottle sizes are 1, 2.25, and 3.8 liters but ACME will sometimes use smaller bottles as a hazard control to limit the deficient reactant (e.g., fuel) that can be released into the combustion chamber.



Imaging Overview

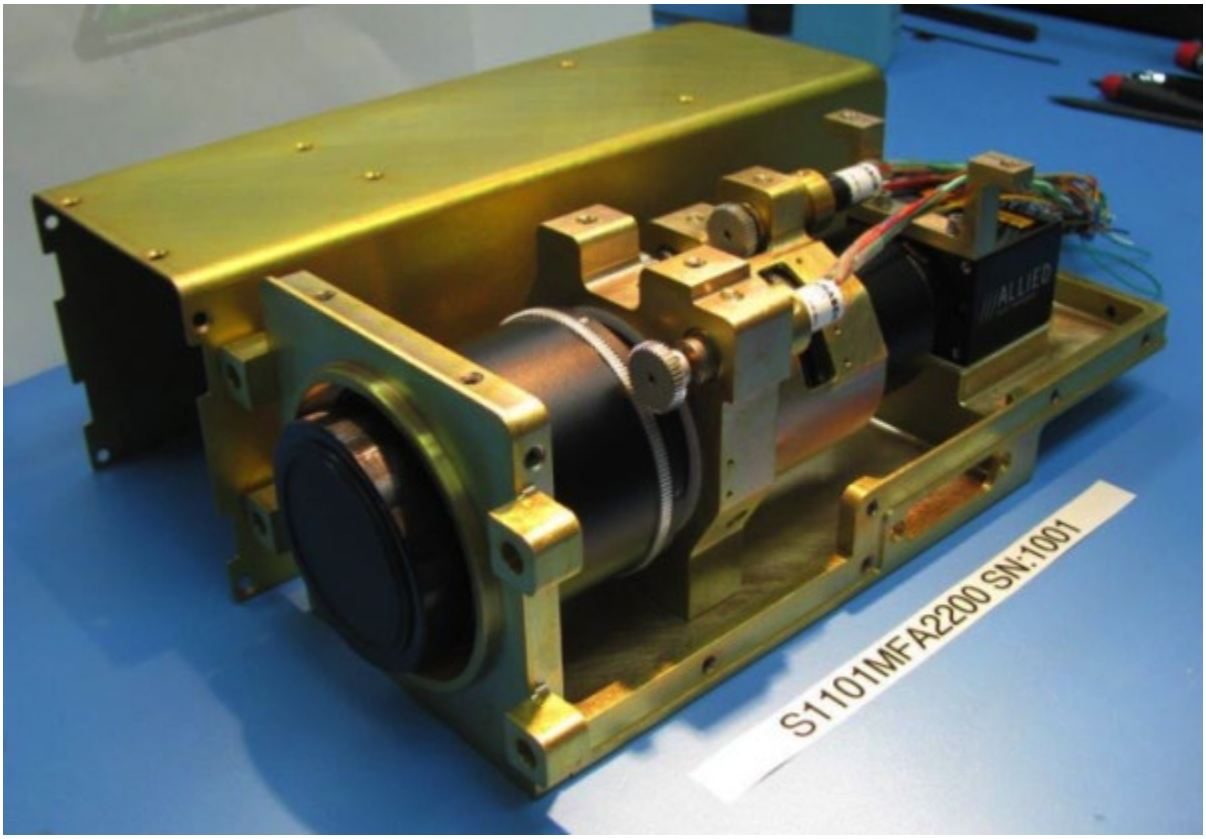
- ACME's primary flame diagnostics are a suite of imaging systems including five video cameras and two illumination systems. Three of the cameras and one illumination system are part of the CIR facility and are already in use on orbit.
- Collectively, the imaging systems will enable color, CH*, and OH* imaging; pyrometry; measurement of the soot volume fraction; and/or operational support.
- All but one of the cameras is mounted outside of the chamber.

Color Imaging

- Color imaging is provided by two video cameras: a high-definition digital camera mounted outside of the chamber and a standard analog camera mounted on the chamber insert (with a turning mirror).
- Both cameras are part of the ACME hardware that is now in development.
- The signal from the analog camera (shown below) will be downlinked in near real time to support the experiment operations, where Light-Emitting Diodes (LEDs) within the chamber can be used to illuminate the igniter and burner to both support the operations and provide reference images.



- The 12-bit digital camera (shown below) records at up to 30 full frames (1360×1024) per second, and is equipped with a motorized lens with independent control of the zoom, iris, and focus.



- The digital camera will be equipped with a filter assembly (shown below) with four options, where selection of a clear “filter” allows for color imaging.
- Color imaging can also be carried out with a blue-green BG7 Schott glass filter, which balances the color planes to avoid saturation in the red while maintaining sensitivity in the blue and green. The natural flame color can be approximately reconstructed, but the primary purpose of the filter is color-ratio pyrometry and measurement of the soot volume fraction.

- Alternately, CH* imaging can be conducted with this camera because the filter assembly includes a 430-nm bandpass filter with a 10-nm full width at half-maximum transmission (FWHM). A 450-nm bandpass filter (again with a 10-nm FWHM) can be used with the CH* filter to effectively subtract interfering broadband soot emission. However, the filter assembly rotates slowly so such subtraction effectively requires a separate test with each filter and of course reproducible flame conditions.



OH* Imaging

- OH* is an excited (radical) molecule that emits light at 310 nm, which is in the ultraviolet, when it drops to a lower energy state in a process known as chemiluminescence.
- ACME will carry out OH* imaging using CIR's Low-Light-Level Ultraviolet (LLL-UV) package, which is a monochrome camera with an intensifier (for 280-700 nm) and a 310-nm bandpass filter (with a 10-nm FWHM) for OH* imaging.

- As a result of the signal loss through the narrow bandpass filter, binning is employed such that the OH* images are recorded at 512×512 and 30 fps.

Color Ratio Pyrometry

- Pyrometry is a technique used to determine an object's temperature from its broadband thermal radiation which is typically infrared but becomes visible at high temperatures. The yellow through red coloring of a typical hydrocarbon flame is caused by the glow of the hot soot particles, whereas the blue is related to the combustion chemistry (and is predominantly CH* emission). Yellow indicates a higher temperature than red, but a ratio of the light emitted at different wavelengths (or wavelength ranges) can be used to determine the temperature, but only where the soot is visibly glowing.
- Pyrometry can be accomplished with CIR's monochrome High-Bit Depth Multispectral (HiBMs) camera, when equipped with a Liquid-Crystal Tunable Filter (LCTF) for imaging at multiple spectra. However, it requires a quasi-steady flame for measurement at each desired wavelength band.
- The HiBMs camera has a telecentric optical system and is recorded at 512×512 and 30 frames per second (fps). Although it has a 1024×1024 pixel array, binning has been necessary because of signal loss through the LCTF filter.
- * For soot-free flames, Thin Filament Pyrometry (TFP) can be used to determine the flame temperature. Ceramic fibers are stretched across the flame and glow when hot, just like the soot particles.
- ACME's chamber insert is equipped with a crew-replaceable array of 5 silicon carbide (SiC) fibers. The TFP fibers are orthogonal to the flame axis and the array can be translated along the flame axis to the desired position downstream of the burner, e.g., based on the downlink of the analog video.
- * During set up, the crew can manually orient the TFP fiber array so that it is orthogonal to either the HiBMs camera with the LCTF or ACME's digital color camera.

Temperature Measurement

- As a supplement to the pyrometry, the ACME chamber insert can accommodate six far-field thermocouples to measure the spread of the thermal field in the spherical flame experiments.
- The thermocouples are of limited utility for the other experiments because they are positioned upstream of the burner tip. Therefore, they will generally not be installed by the crew to reduce the chance of accidental damage.
- There will be two versions of a four-thermocouple rake to allow a choice of near and far measurements. Other arrays could be designed to support future experiments, but they cannot extend downstream of the burner tip without interfering with other (e.g., optical) measurements.

Soot Volume Fraction

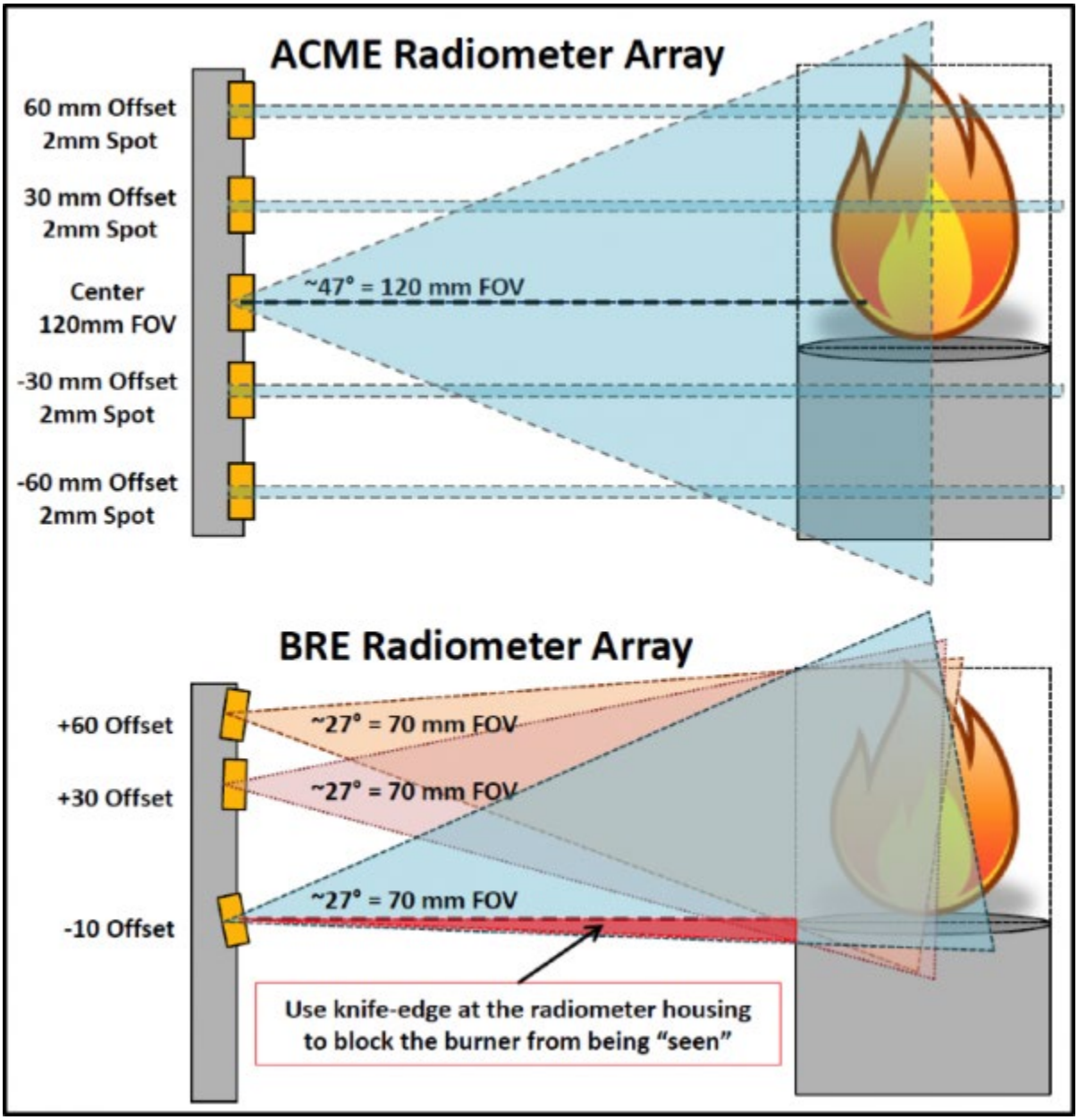
- The amount of soot particulate in a flame can be determined via an extinction technique, where the camera and the collimated light source are positioned at opposite windows.
- For ACME, the extinction measurement will be made with a second High-Bit Depth Multispectral (HiBMs) camera, without a filter, in combination with CIR's illumination package (with its two fiber-coupled laser diodes). The images are recorded at 1024×1024 and 30 frames per second.
- Alternatively, the soot volume fraction can be determined directly from the digital color camera, where an alternate Thin Filament Pyrometry (TFP) fiber array including a platinum wire will be used for inflight characterization for that measurement.

Thermal Radiation

- The ACME chamber insert includes a crew-exchangeable linear array of radiometers of which there are two versions supporting the initial set of experiments. Alternate arrays could be developed for future experiments.



- The radiometer array has a fixed position which is parallel to the chamber and burner axis and is centered (along the chamber length) relative to the chamber windows and thus most of the other instrumentation.



- An additional wide-view radiometer, that is fixed and not crew exchangeable, is positioned downstream of the burner and flame at 30o from the chamber axis. This detector is primarily intended to support the spherical flame experiments and will have a quicker response than the more sensitive detectors that will used in the linear arrays.
- ACME's radiometers are more precisely thermopile detectors with a spectral range of nominally 0.2-11 micron and fields of view that are established by appropriately selected apertures (i.e., rather than lenses).

Chemiluminescent Emission

- To detect the quenching of dim flames, the chamber insert will be equipped with a set of three Photomultiplier Tube (PMT) modules.

- Each PMT will have a wide-angle view, a sight line that is orthogonal to the chamber axis, and will be centered (along the chamber length) relative to the chamber windows so that it is aligned with most of the other instrumentation.
- The PMTs' spectral range will be 230-700 nm, i.e., visible and near ultraviolet, but one will have a 310-nm bandpass filter and another will have a 430-nm bandpass filter (where both are with a 10-nm FWHM). The filtered pair of detectors will enable measurement of OH* and CH* emission, respectively. The third PMT will be equipped with a neutral density filter to prevent saturation. Like many other ACME components, the PMT array can be removed from the chamber insert by the crew.

Other Measurements

- Additional supporting measurements will be made for the ACME experiments, where the chamber pressure is a simple example.
- The ACME experiments will use the CIR's gas chromatograph for pre- and post-test analysis of the chamber atmosphere, e.g., for oxygen, nitrogen, carbon dioxide, carbon monoxide, and fuel gases.
- Acceleration measurements will be provided by the Space Acceleration Measurement System II (SAMS-II), which is the primary instrument for such measurements on ISS. Its predecessor previously served that role for the space shuttle.

Experiment-Unique Equipment

- Experiment-unique hardware is also mounted on the chamber insert. For example, to accommodate the E-FIELD Flames experiment, the insert is equipped with a high-voltage supply and a crew-removable electrode mesh which can be set between +10 and -10 kV. The circular mesh is orthogonal to the chamber axis and is positioned (during set up) at either ~35 or ~50 mm downstream of the insulated burner. The ion current between the burner and electrode mesh will be determined via measurement of the voltage drop across a known resistor and application of Ohm's Law.

ACME Operations

- All ACME experiments will be constant volume (i.e., rather than constant pressure) studies, where the chamber's free volume will be about 90 liters. Burner flows will generally be constrained to limit the pressure rise, and tests will typically be 0.5-2 minutes in duration.
- The ACME experiments will be configured, but not conducted, by the ISS crew. Instead, the experiments will be commanded from the ground, specifically from the Telescience Support Center (TSC) at the NASA Glenn Research Center.
- The ACME tests can be conducted in a nominally automated mode using pre-programmed scripts or in a "manual" mode where changes can be made (e.g., in the burner flow) in response to the analog video downlink and/or the numerical data available in the telemetry stream. The latter mode can facilitate the conduct of tests at or near limit conditions, e.g., for soot or stability.

Gallery

Need to upload from previous site.

Videos

Need to upload from previous site.

ACME Contact Information

Project Manager: Lauren Brown, NASA Glenn Research Center, 216-433-8429, lauren.brown@nasa.gov

Project Scientist: Dennis Stocker, NASA Glenn Research Center, 216-433-2166, Dennis.P.Stocker@nasa.gov

Deputy Project Scientist: Prof. Fumiaki Takahashi, Case Western Reserve University, 216-368-6838, fumiaki.takahashi-1@nasa.gov

Engineering Team: ZIN Technologies, Inc.